

**ADMINISTRATIVE INFORMATION****FMI No: 25326****FMI Title: DAS Fuse Replacement for Ltspd 16**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety ☐ Mandatory ☐ Mandatory on Request	GEMS-AM: GEMS-E: GEMS-A:	Issue Date: Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole. ☒ Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole. ☐ Parts Ordering: GEMS - Americas, contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator ☐ No Parts are required to complete this FMI.
☐ Batch with	FMI # To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked:** LightSpeed 16 Gantries: 2339985, 2339985-2**FMI Instructions Part No.: 2375792-100****Do NOT leave unsecured on site.**



Completion Certification Report

Mail To

GEMS - Americas

GE Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe

GEMS - E / ESC
FMI Admin. 322
BP34
F 78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25326 FMI Title: DAS Fuse Replacement for Ltspd 16

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code:

- 1.) Modification Complete or previously modified.
- 2.) FMI not required per FMI instruction.
- 3.) Not modified. Unit in storage. FMI kit returned to stock.
- 4.) Modification refused or equipment altered. Customer signature and title required.
- 5.) Unit location unknown or unit not in area. Product Locator notified.

**Customer Refusal or Equipment Altered
- Modification Pending.**

Installer
Comments

Effectivity

The following CT Equipment is affected:

2339985 LightSpeed 16 Gantry Assembly

2339985-2 LightSpeed 16 Gantry Assembly

Purpose

The purpose of this FMI is to replace an under rated fuse holder. This fuse holder overheats and causes the fuse within to fail.

Furnished Materials

DAS Fuse Replacement Kit - FMI 25326 (Part # 2378460)

Qty	Description	Part #
1	FMI Document	2375792-100
1	Fuse Holder, 30 Amp 250V	46-170042P28
3	Fuse, 20 Amp 250 Volt, 3AB Slow-Blow	46-170021P81
1	Fuse box switch wire	2378459
8	Pop Rivets (4 extra)	46-220186P20
1	Hand Rivet Gun	2384121
1	3mm drill bit	2387528

Required Tools

Qty	Description	Part #
1	FMI Document	2375792-100
1	45 watt Soldering iron	Field Tool
1	Standard 60/40 Rosin core solder	Field Supplied
1	Digital Volt Meter	Field Supplied
1	Laptop	Field Supplied
1	LightSpeed Service CD #2360027-200	Field Supplied
1	Drill Motor	Field Supplied

People Power

2 hours labor and 1 hour of travel

Procedure

1 Preparation *or* Before You Begin

INSPECTION

- 1.) Remove gantry right side cover, disable axial drive and rotate gantry by hand to see DAS fuse box. See [Figure 2](#) to help identify DAS fuse box.
- 2.) Inspect Fuse Box Assembly: Is **DAS** Power Supply Fuse Holder Grey in Color?
 - a.) Yes: You are done. FMI is not needed. Use **FMI** completion Code 2 and return unused kit to: HM Product Solutions, Ltd., 3225 Gateway Road, Suite 250, Brookfield, WI 53045.
 - b.) No: Proceed to Part 2 and complete **FMI**.
- 3.) Run DASTOOLs Auxiliary channels and record power supply voltages. This will be your baseline.

2 HW Adjustment and Installation

- 1.) Plug in soldering iron so it is ready for use when needed.
- 2.) Remove both side and top covers. Disable gantry axial drive using service switch.
- 3.) Turn off **HVDC** and 120VAC service switches. Turn off **DAS** power on the **DAS** fuse box.
- 4.) Remove back gantry cover, plugging in estop cable to terminator so you can enable the gantry for tilt.
- 5.) Turn 120VAC back on, reset scan hardware via SERVICE DESKTOP - SYSTEM RESETS - SCAN RESET selection.
- 6.) Raise table to the max position to allow the gantry to tilt back 30 degrees.
- 7.) With Axial drive disabled, position rotating frame with fuse holder on top (near 12 o'clock), on either side of center top cover support. Engage gantry rotation lock.
- 8.) Tilt gantry back full 30 degrees as shown in [Figure 1](#) below. Turn off 120VAC service switch. Verify DAS power is still off.



Figure 1 - Gantry in position for FMI procedure

- 9.) Unplug J99 from center **DAS** backplane & J's 1, 2, & 3 from **DHCB** (Detector Heater Control Board) to gain easier access.
- 10.) Loosen 3 slot screws and remove two holding screws from fuse holder, which will slide down for removal. See [Figure 2](#) below.

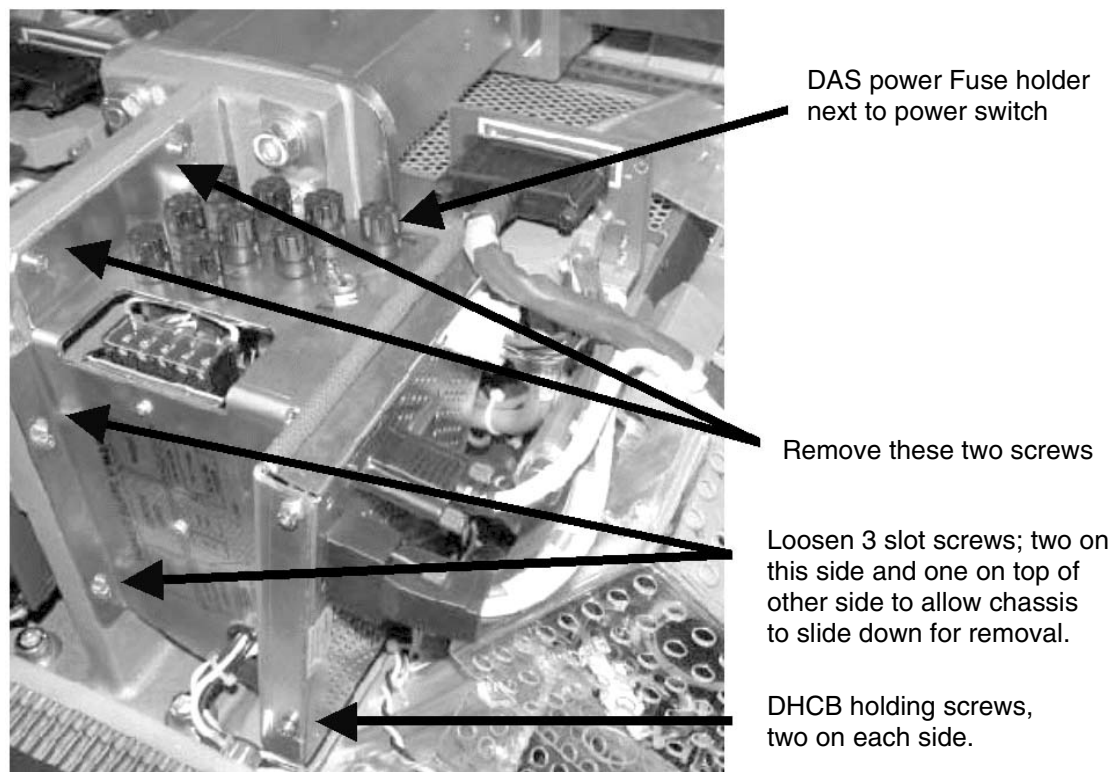


Figure 2 - Component Location Diagram

**DANGER**

- 11.) Verify no power present in fuse box with your **DVM**.
- 12.) Loosen the lower 2 **DHCB** holding screws (slotted) & remove the upper 2 holding screws, be careful there are two washers on each. Remove **DHCB**.
- 13.) Position covers (towels, sheet, etc.) under fuse block to keep metal filings out of the gantry. Using the supplied drill bit to drill out the 4 pop rivet heads that are holding the perforated panel behind the **DHCB**. There are 2 rivets on the side panel below the **DAS** Fuse holder and 2 on the face of the perforated panel on the opposite side. Use a vacuum hose set near drilling location while drilling to help capture filings.
- 14.) After drilling rivet head off, use a small punch to push the remainder of rivet into fuse holder box on front side; you then must remove these also. Punch out the rivet studs in the holes, catching them in your hand. The towel or drape under the assembly can help catch these.
- 15.) Perforated panel can now be removed, be careful of grid/filter tape or Ty-Raps that may be holding wires to perforated panel. Cut Ty-Raps if present. The **DAS** power fuse and switch are now accessible as shown in [Figure 3](#) below.
- 16.) Remove wires to existing **DAS** power fuse holder; remove existing fuse holder.
- 17.) Remove old fuse to switch wire at the switch. Cut Ty-Raps carefully as needed.
- 18.) Install new fuse holder, making sure to position terminal 1 connector toward the open side for ease of soldering wires on. Follow [Figure 4](#) for installing fuseholder parts.

19.) Solder new wire (supplied with kit) to terminal 1 of new fuse holder (see Figure 4 below).

Note: You **must** use new larger gauge wire supplied for the Fuse holder to Switch connection. The rest of the harness is already larger gauge wire. Failure to do so may result in wire insulation melting or degrading over time.

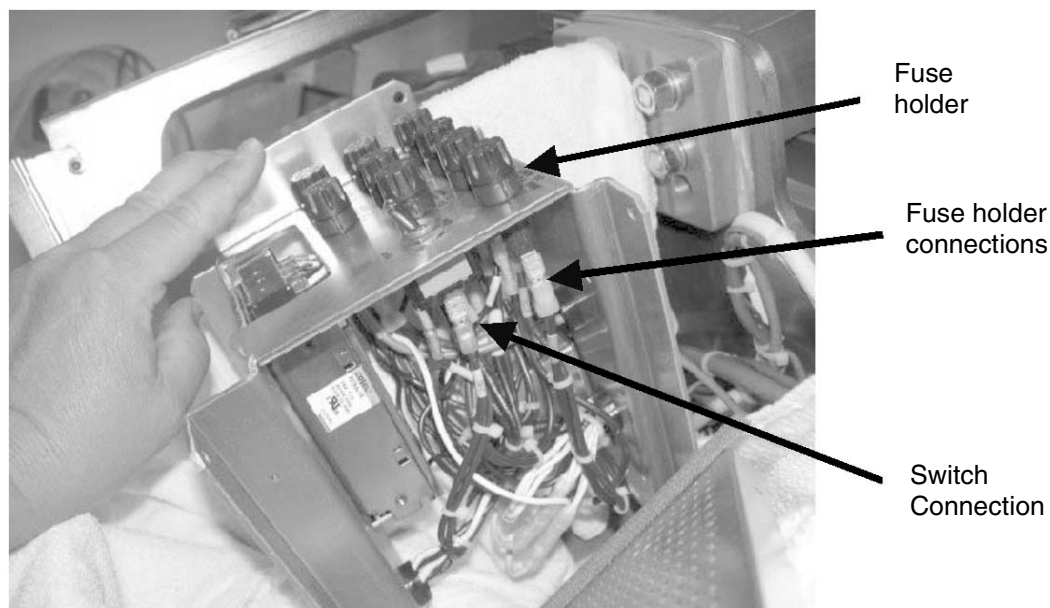


Figure 3 - Fuse Box with screen cover removed

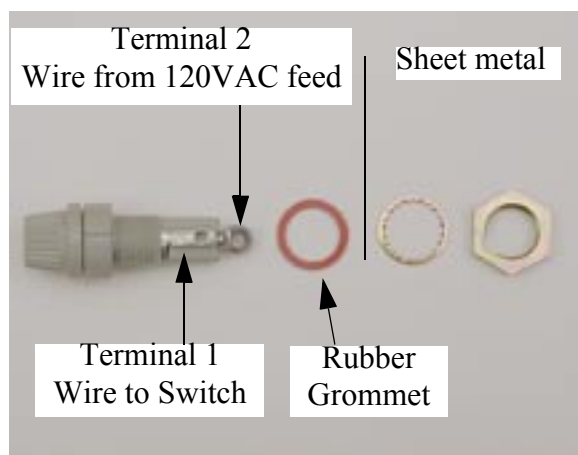


Figure 4 - New 30 Amp Rated Fuse Holder

- 20.) Cut off spade lug from the terminal 2 wire (F6-c) as close to lugs as possible. Strip back wire 13mm to allow for soldering. Wires are labeled f6-s (side) & f6-c (center).
- 21.) Tin ends and solder to fuse holder connections. Check solder joints carefully; you don't want to have to come back here! It is not necessary to place wires through holes of fuse holder terminals.
- 22.) Position wires and check clearance to perforated plate to be reinstalled.
- 23.) Use the supplied Pop Rivets and Rivet gun to reinstall the perforated plate to the **DAS** fuse holder chassis. Make sure to install the pop rivets in the orientation show in [Figure 5](#) below. Failure to do so can cause interference with the **DHCB** for two of the pop rivets when the **DHCB** is reinstalled.

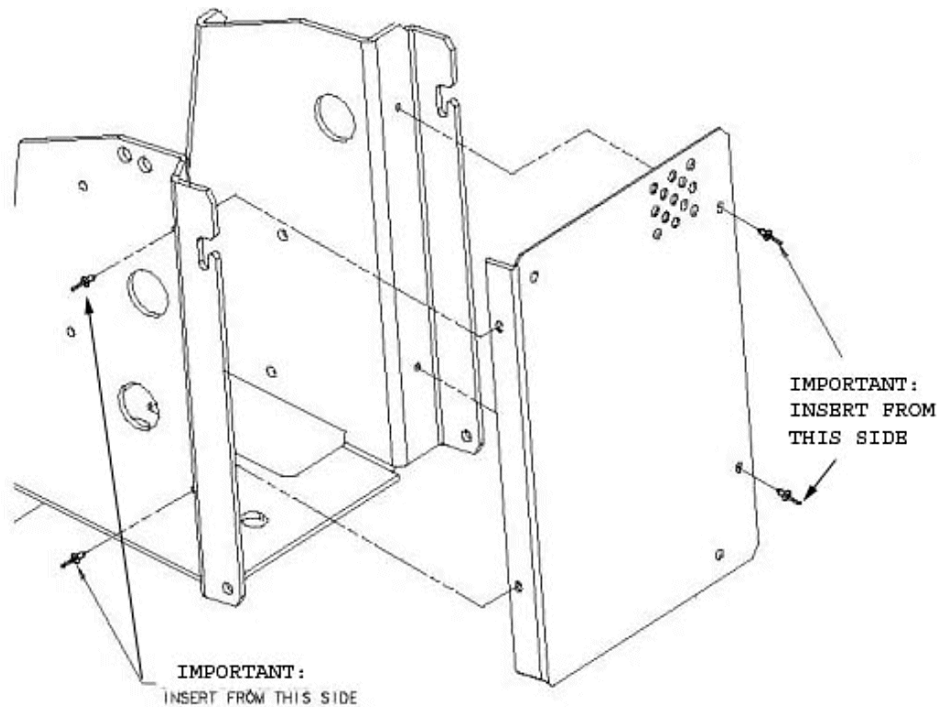


Figure 5 - Pop Rivet Insertion

- 24.) Carefully remove covering under fuse block and vacuum / clean any debris.
- 25.) Reinstall **DHCB** using screws and washers removed earlier and tighten slot screws.
- 26.) Reinstall fuse block being careful to position blk/wht wire harness coming out back side so it goes down below fuse block under the large round hex head bolt. This ensures clearance from belt and gear hardware during rotation.
- 27.) Install new fuse supplied with kit.
- 28.) Reinstall **DAS** cable and **DHCB** cables. Check clearance of all cables.

3 Finishing Up

- 1.) Restore power to gantry by turning on DAS power switch on the DAS fuse holder chassis and turning on all 3 service switches (120V, HVDC, Axial drive enable).
- 2.) Allow system to warm up for 30 minutes prior to running final IQ tests in step 6 to allow **DAS** to reach proper temperature.
- 3.) Run DASTOOLS Auxiliary Channels.
- 4.) Check power supplies using [Figure 6](#) for test points and adjust as necessary.

Note: Use Service CD 2360027-200, Direction 2341960, System Service Manual (Adv) Rev 1, Chapter 8 Gantry, Section 2.1.4 to adjust power supply levels if necessary. Do not adjust supplies unless they are out of spec. DO NOT ATTEMPT TO ADJUST THESE SUPPLIES WITHOUT USING THE SERVICE MANUAL.

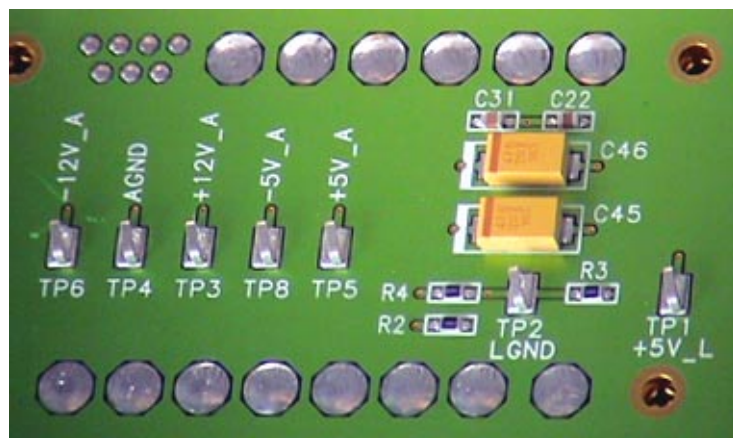


Figure 6 - DAS Center Chassis Voltage Adjustment Test points

- 5.) Install gantry covers.
- 6.) Perform Image Quality test scans.
 - a.) Place the QA phantom on the phantom holder.
 - I.) Drive the table to an elevation of 100.
 - II.) Align the reference line on the phantom with the internal laser lights.



NOTICE
Equipment
Damage
Possible

Avoid Detector Damage. Never scan above 50 mA without first placing a phantom in the field of view. Levels in excess of 50 mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

- b.) Manually select the scan parameters in [Table 1](#).
- c.) Complete the scans.

Scan Type	Kv	mA	SFOV	Thickness	Scan Time	Start Location	End Location	Tilt/Pitch
Scout	120	40				S200	I800	0, HQ
Scout	120	40				S200	I800	90, HS
Cine	120	50	Large	4x5	30sec	S0	S0	1.0, HQ
Axial	120	50	Large	4x5	1.0sec	S0	S0	I30, HS
Axial	120	50	Large	4x1.25	1.0sec	S0	S0	S30, HQ
Helical	120	50	Large	4x1.25	30sec	S70	I75	HS

Table 1 - System Scanning Scan Test Parameters

